IMPLICIT DIFFERENTIATION

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Name: Solutions

Use implicit differentiation to find $\mathrm{d}y/\mathrm{d}x$.

1. $2x^2 + xy - y^2 = 2$

$$\frac{d}{dx}(2x^2+xy-y^2) = 4x + (y + xy') - 2yy' = 0$$

$$\Rightarrow 4x + y = 2yy' - xy' = (2y - x)y'$$

$$\Rightarrow y' = \frac{dy}{dx} = \boxed{\frac{4x+y}{2y-x}}.$$

2. $x^3 - xy^2 + y^3 = 1$

$$0 = 3x^2 - (y^2 + x(2yy')) + 3y^2y'$$

$$= 3x^2 - y^2 - 2xyy' + 3y^2y'$$

$$\Rightarrow y^2 - 3x^2 = (-2xy + 3y^2)y'$$

$$\Rightarrow \boxed{y' = \frac{y^2-3x^2}{3y^2-2xy}}$$

3. $y^5 + x^2y^3 = 1 + x^4y$

$$5y^4y' + (2xy^3 + x^2(3y^2y')) = 0 + 4x^3y + x^4y'$$

$$\Rightarrow 5y^4y' + 3x^2y^2y' + 2xy^3 = 4x^3y + x^4y'$$

$$\Rightarrow 5y^4y' + 3x^2y^2y' - x^4y' = y'(5y^4 + 3x^2y^2 - x^4) = 4x^3y - 2xy^3$$

$$\Rightarrow y' = \boxed{\frac{4x^3y - 2xy^3}{5y^4 + 3x^2y^2 - x^4}}$$

4. $\cos(xy) = 1 + \sin(y)$

$$-\sin(xy)(y + xy') = 0 + \cos(y)y' = \cos(y)y'$$

$$\Rightarrow -y\sin(xy) - xy'\sin(xy) = \cos(y)y'$$

$$\Rightarrow -y\sin(xy) = \cos(y)y' - xy'\sin(xy) = (\cos(y) + x\sin(xy))y'$$

$$\Rightarrow y' = \boxed{\frac{-y\sin(xy)}{\cos(y) + x\sin(xy)}}$$

5. $y\sin(x^2) = x\sin(y^2)$

$$\frac{d}{dx} y\sin(x^2) = y'\sin(x^2) + y\cos(x^2)(2x) = y'\sin(x^2) + 2xy\cos(x^2)$$

$$\frac{d}{dx}(x\sin(y^2)) = \sin(y^2) + x\cos(y^2)\frac{d}{dx}(y^2) = \sin(y^2) + 2xyy'\cos(y^2)$$

$$\Rightarrow y'\sin(x^2) + 2xy\cos(x^2) = \sin(y^2) + 2xyy'\cos(y^2)$$

$$\Rightarrow y'\sin(x^2) - 2xyy'\cos(y^2) = \sin(y^2) - 2xy\cos(x^2)$$

$$\Rightarrow y'(\sin(x^2) - 2xy\cos(y^2)) = \sin(y^2) - 2xy\cos(x^2)$$

$$\Rightarrow y' = \boxed{\frac{\sin(y^2) - 2xy\cos(x^2)}{\sin(x^2) - 2xy\cos(y^2)}}$$

6. $xy = \sqrt{x^2 + y^2}$

$$\Rightarrow x^2y^2 = x^2 + y^2$$

$$\Rightarrow 2xy^2 + x^2(2yy') = 2x + 2yy'$$

$$\Rightarrow 2xy^2 - 2x = 2yy' - 2x^2yy' = (2y - 2x^2y)y'$$

$$\Rightarrow y' = \frac{2xy^2 - 2x}{2y - 2x^2y} = \boxed{\frac{xy^2 - x}{y - x^2y}}$$

7. $x\sin(y) + y\sin(x) = 1$

$$0 = \sin(y) + x\cos(y)y' + y'\sin(x) + y\cos(x)$$

$$\Rightarrow \sin(y) + y\cos(x) = -x\cos(y)y' - \sin(x)y' = -(x\cos(y) + \sin(x))y'$$

$$\Rightarrow -y'(x\cos(y) + \sin(x)) = y\cos(x) + \sin(y)$$

$$\Rightarrow y' = \boxed{\frac{-y\cos(x) - \sin(y)}{x\cos(y) + \sin(x)}}$$

8. $\tan(x - y) = \dfrac{y}{1 + x^2}$

$$\frac{d}{dx}\tan(x-y) = \sec^2(x-y)(1-y') \; ; \quad \frac{d}{dx}\frac{y}{1+x^2} = \frac{y'(1+x^2) - y(2x)}{(1+x^2)^2}$$

$$\Rightarrow \sec^2(x-y) - y'\sec^2(x-y) = \frac{y'(1+x^2) - 2xy}{(1+x^2)^2} = \frac{y'}{1+x^2} - \frac{2xy}{(1+x^2)^2}$$

$$\Rightarrow \sec^2(x-y) + \frac{2xy}{(1+x^2)^2} = \frac{y'}{1+x^2} + y'\sec^2(x-y)$$

$$\Rightarrow \sec^2(x-y)(1+x^2)^2 + 2xy = y'(1+x^2) + y'\sec^2(x-y)(1+x^2)^2 \Rightarrow y' = \boxed{\frac{\sec^2(x-y)(1+x^2)^2 + 2xy}{(1+x^2) + \sec^2(x-y)(1+x^2)^2}}$$

9. You are given that $f(1) = 2$ and $f(x) + x^2f(x)^3 = 10$. Find $f'(1)$.

$$f'(x) + 2xf(x)^3 + 3x^2f(x)^2f'(x) = 0$$

$$\Rightarrow f'(x)(1 + 3x^2f(x)^2) = -2xf(x)^3$$

$$\Rightarrow f'(x) = \frac{-2xf(x)^3}{1 + 3x^2f(x)^2}$$

$$f'(1) = \frac{-2(1)f(1)^3}{1 + 3(1)^2f(1)^2} = \frac{-2(2)^3}{1 + 3(2)^2} = \boxed{\frac{-16}{13}}$$